ALEX LI

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TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, SQL, Java, JavaScript, SystemVerilog, HTML/CSS
- **Frameworks/Libraries:** FastAPI, Spring Boot, Express, React, Next, PyTorch, NumPy, pandas
- **Tools & Platforms:** Linux, Git, Make, PostgreSQL, Terraform, Docker, FPGAs, AWS, gprof, gRPC

EDUCATION

University of Washington, Seattle, WA — Expected June 2027
B.S., Computer Science

Relevant Coursework: Discrete Mathematics, Linear Algebra, Data Structures and Parallelism, Hardware/Software Interface, Systems Programming, Digital Circuit Design, Machine Learning, Distributed Systems, Data Management, Datacenter Systems, Compilers
Awards: Best Use of AI + Best Finance/Data Analytics Project (HackTech 2025), **1st Place** DubHacks 2023 (Synergy Track), **Futureforce Selectee** (Salesforce Data Cloud), Dean's List (2023–25)

RELEVANT EXPERIENCE

Copperlane (YC W26) | *AWS (DynamoDB, SQS, S3), FastAPI, Pydantic* — November 2025 – May 2026
Founding Engineer / Co-founder

- Designed and shipped core backend systems for a **multi-tenant B2B SaaS platform**, including tenant-scoped APIs, DynamoDB data models, async workflow execution, and agent orchestration powering early enterprise customers and **high five-figure contracted ARR**.
- Built an **SQS FIFO-based workflow queue and retry system** with message-group ordering, deduplication IDs, idempotent at-least-once processing, DLQs, and idempotency records to make notifications and audit writes **safe across redeliveries**.

Chewy | *Spring Boot, Terraform, AWS (DynamoDB, SQS, Lambda), Mockito, Guice* — June 2025 – August 2025
Software Engineering Intern

- Built and deployed a **backend microservice** powering Chewy Ads metrics reporting, replacing a 3rd-party scheduler with an **in-house job orchestration system** using AWS Lambda, SQS, and DynamoDB.
- Designed **retry-safe distributed workflows** with exponential backoff, DLQs, and SHA-256 deduplication.

UW Sensor Systems Lab | *FPGAs, Verilog, GTKWave, Python, Serial Interfaces* — December 2024 – June 2025
Undergraduate Research Assistant

- Developed **low-level FPGA drivers** and replaced I^2C with a UART protocol, increasing bandwidth and throughput for acoustic levitation control.
- Implemented **real-time synchronization** logic across 100+ ultrasonic transducers for deterministic phase-angle coordination in object manipulation experiments.

PROJECTS

Othello Game Engine | *C++23, Multithreading, Profiling, gRPC, Docker* — GitHub

- Engineered a high-performance **parallel game engine** using iterative-deepening α-β search with principal variation search (PVS) and Zobrist-hashed transposition tables.
- **Optimized multithreaded throughput** to 18.5M nodes/s by sharding the transposition table per thread and eliminating STL overhead in the move-generation hot path.
- Exposed engine functionality through a **gRPC microservice API** supporting real-time remote play and orchestration; containerized with Docker for reproducible deployment.

Distributed Key-Value Store (Multi-Paxos, Sharded) | *Java, Multi-Paxos, Sharding*

- Designed and implemented a **fault-tolerant distributed key-value store** with leader election, replication, and dynamic shard rebalancing using the Multi-Paxos consensus algorithm.
- Built consensus-driven reconfiguration logic ensuring **linearizable consistency** under network partitions and crash faults; validated with adversarial distributed tests.